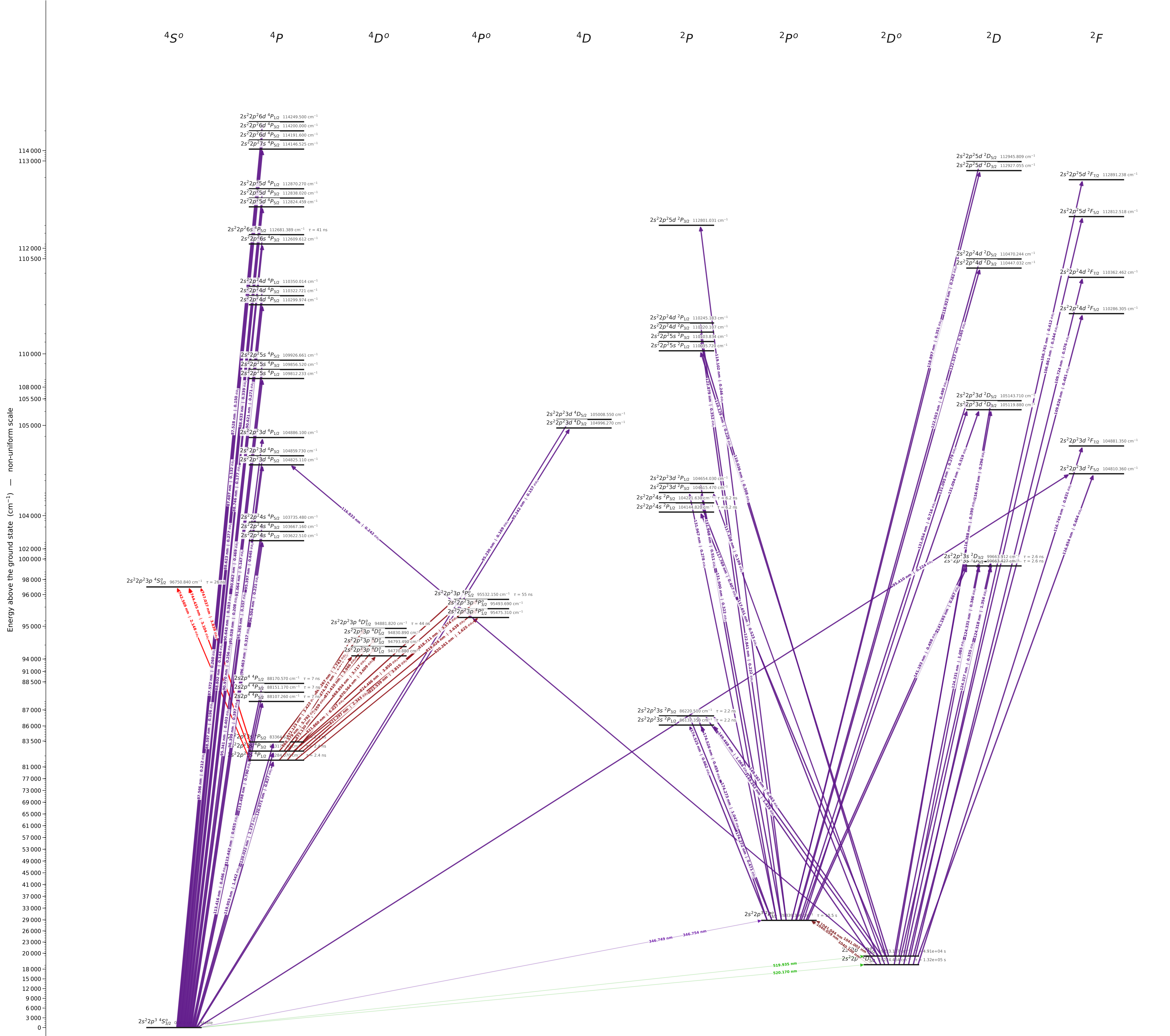
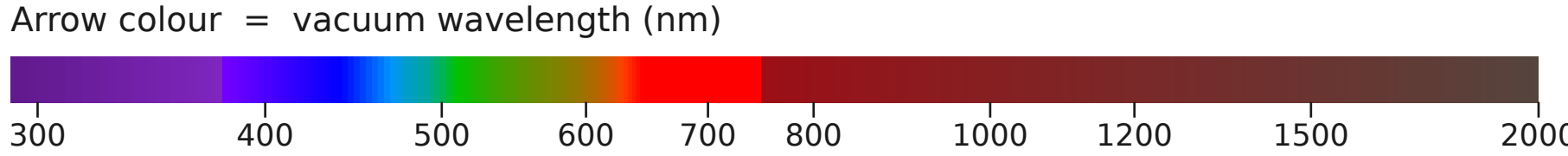


# Nitrogen

N (N I). Strongest classified transitions below 2 μm (NIST): 67 levels, 96 lines.

N<sup>+</sup> ionisation limit 117225.70 cm<sup>-1</sup> = 14.53413 eV (85.306 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|U||e r||J\rangle|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                              | λ vac (nm) | λ air (nm) | ν (THz)  | $ U  e r  J\rangle $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                |
|---------------------------------------------------------|------------|------------|----------|-----------------------------------------|----------------------|----------------|------------|----------------------------------------------------|
| 2s²2p³ 4S <sub>3/2</sub> - 2s²2p³ ²P <sub>3/2</sub>     | 346.7490   | 346.6497   | 864.5806 | -                                       | 6.500e-03            | 1.511e-08      | 6.84       | auroral [N I] 346.6 nm line                        |
| 2s²2p³ 4S <sub>3/2</sub> - 2s²2p³ ²P <sub>1/2</sub>     | 346.7536   | 346.6543   | 864.5691 | -                                       | 2.600e-03            | 1.436e-08      | 2.88       | auroral [N I] 346.6 nm line                        |
| 2s²2p³ 4S <sub>3/2</sub> - 2s²2p³ ²D <sub>3/2</sub>     | 519.9349   | 519.7902   | 576.5961 | -                                       | 1.600e-05            | 3.239e-12      | 78.6       | auroral / nightglow / nebular [N I] 520 nm doublet |
| 2s²2p³ 4S <sub>3/2</sub> - 2s²2p³ ²D <sub>5/2</sub>     | 520.1705   | 520.0257   | 576.3349 | -                                       | 6.590e-06            | 1.203e-12      | 87.2       | auroral / nightglow / nebular [N I] 520 nm doublet |
| 2s²2p³3s 4P <sub>1/2</sub> - 2s²2p³3p 4S <sub>1/2</sub> | 742.5686   | 742.3641   | 403.7236 | 2.148                                   | 5.710e+06            | 6.121          | 14.8       | 742.57 nm                                          |
| 2s²2p³3s 4P <sub>3/2</sub> - 2s²2p³3p 4S <sub>3/2</sub> | 744.4348   | 744.2298   | 402.7115 | 3.104                                   | 1.183e+07            | 6.121          | 30.8       | 744.43 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4S <sub>5/2</sub> | 747.0369   | 746.8312   | 401.3088 | 3.839                                   | 1.791e+07            | 6.121          | 46.6       | 747.04 nm                                          |
| 2s²2p³3s 4P <sub>3/2</sub> - 2s²2p³3p 4P <sub>1/2</sub> | 818.7112   | 818.4861   | 366.1761 | 3.574                                   | 7.860e+06            | 2.894          | 43.2       | 818.71 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4P <sub>3/2</sub> | 819.0263   | 818.8012   | 366.0352 | 3.638                                   | 1.220e+07            | -              | -          | 819.03 nm                                          |
| 2s²2p³3s 4P <sub>1/2</sub> - 2s²2p³3p 4P <sub>1/2</sub> | 820.2611   | 820.0357   | 365.4842 | 1.625                                   | 4.850e+06            | -              | -          | 820.26 nm                                          |
| 2s²2p³3s 4P <sub>3/2</sub> - 2s²2p³3p 4P <sub>3/2</sub> | 821.2972   | 821.0715   | 365.0231 | 2.343                                   | 5.020e+06            | -              | -          | 821.30 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4P <sub>5/2</sub> | 821.8595   | 821.6336   | 364.7734 | 6.027                                   | 2.210e+07            | 2.894          | 100        | 821.86 nm                                          |
| 2s²2p³3s 4P <sub>3/2</sub> - 2s²2p³3p 4P <sub>1/2</sub> | 822.5389   | 822.3128   | 364.4721 | 3.815                                   | 2.650e+07            | -              | -          | 822.54 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4P <sub>3/2</sub> | 824.4655   | 824.2389   | 363.6204 | 3.850                                   | 1.340e+07            | -              | -          | 824.47 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>1/2</sub> | 868.2666   | 868.0282   | 345.2770 | 7.743                                   | 2.320e+07            | 3.617          | 100        | 868.27 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>3/2</sub> | 868.5788   | 868.3403   | 345.1529 | 5.792                                   | 1.729e+07            | -              | -          | 868.58 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>5/2</sub> | 868.8535   | 868.6149   | 345.0437 | 3.717                                   | 1.067e+07            | -              | -          | 868.85 nm                                          |
| 2s²2p³3s 4P <sub>1/2</sub> - 2s²2p³3p 4D <sub>1/2</sub> | 870.5637   | 870.3247   | 344.3659 | 3.609                                   | 2.000e+07            | -              | -          | 870.56 nm                                          |
| 2s²2p³3s 4P <sub>3/2</sub> - 2s²2p³3p 4D <sub>3/2</sub> | 871.4096   | 871.1703   | 344.0316 | 3.966                                   | 1.204e+07            | -              | -          | 871.41 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>5/2</sub> | 872.1232   | 871.8837   | 343.7501 | 3.422                                   | 5.960e+06            | -              | -          | 872.12 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>1/2</sub> | 873.1299   | 872.8901   | 343.3538 | 1.559                                   | 3.700e+06            | -              | -          | 873.13 nm                                          |
| 2s²2p³3s 4P <sub>5/2</sub> - 2s²2p³3p 4D <sub>3/2</sub> | 874.9771   | 874.7369   | 342.6289 | 1.079                                   | 8.800e+05            | -              | -          | 874.98 nm                                          |
| 2s²2p³ ²D <sub>3/2</sub> - 2s²2p³ ²P <sub>3/2</sub>     | 1040.0587  | 1039.7737  | 288.2457 | -                                       | 6.000e-02            | 1.511e-08      | 63.2       | [N I] 1040 nm infrared auroral multiplet           |
| 2s²2p³ ²D <sub>5/2</sub> - 2s²2p³ ²P <sub>5/2</sub>     | 1040.1004  | 1039.8155  | 288.2341 | -                                       | 3.450e-02            | 1.436e-08      | 38.2       | [N I] 1040 nm infrared auroral multiplet           |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πc).

## <sup>14</sup>N hyperfine structure (*I* = 1)

| Level                      | A (MHz)   | B (MHz)  | Shift of each F level from the centre of gravity (MHz) |
|----------------------------|-----------|----------|--------------------------------------------------------|
| 2s²2p³ 4S <sub>3/2</sub>   | 10.450929 | 0.000001 | F=1/2: -26.13 F=3/2: -10.45 F=5/2: +15.68              |
| 2s²2p³3s 4P <sub>1/2</sub> | 112.3(13) | -        | - F=1/2: -112.30 F=3/2: +56.15                         |
| 2s²2p³3s 4P <sub>3/2</sub> | 68.3(7)   | 4(9)     | F=1/2: -166.45 F=3/2: -71.83 F=5/2: +103.37            |
| 2s²2p³3s 4P <sub>5/2</sub> | 129.5(8)  | -7.8(20) | F=3/2: -458.78 F=5/2: -123.28 F=7/2: +321.85           |
| 2s²2p³3p 4D <sub>1/2</sub> | 69.8(9)   | -        | - F=1/2: -69.76 F=3/2: +34.88                          |
| 2s²2p³3p 4D <sub>3/2</sub> | 30.3(15)  | -0.9(17) | F=1/2: -76.88 F=3/2: -29.40 F=5/2: +45.23              |
| 2s²2p³3p 4D <sub>5/2</sub> | 36.6(11)  | -4.1(25) | F=3/2: -130.97 F=5/2: -33.32 F=7/2: +90.47             |
| 2s²2p³3p 4D <sub>7/2</sub> | 55.2(11)  | -9.9(26) | F=5/2: -253.70 F=7/2: -48.13 F=9/2: +190.72            |
| 2s²2p³3p 4P <sub>1/2</sub> | -50.8(17) | -        | F=1/2: +50.78 F=3/2: -25.39                            |
| 2s²2p³3p 4P <sub>3/2</sub> | 46.2(15)  | -2.7(17) | F=1/2: -118.88 F=3/2: -43.50 F=5/2: +68.62             |
| 2s²2p³3p 4P <sub>5/2</sub> | 31.9(9)   | 1.1(21)  | F=3/2: -110.98 F=5/2: -32.81 F=7/2: +80.10             |

Measured A, B constants; sources in the data file.